

Weekly CT - Chapters 01-03 | Answer Sheet

Cambridge International AS and A Level Computer Science 9618

Data representation | Networks | Hardware and memory

Time: 30 minutes

Main marks: 29

Bonus: 1 (Total: 30)

Instructions: Award marks for equivalent wording that communicates the same technical point. Calculations should show correct method where indicated.

1. Run-length encoding (RLE)

[3]

- (a) 5R 2B 6G 3R. Accept an equivalent unambiguous count/value representation. [2]
(b) The original pixel sequence can be reconstructed exactly; no data is discarded. [1]

2. Image representation and file size

[3]

- (a) 24 bits are used to represent the colour of each pixel (allowing up to 2^{24} different colour values). [1]
(b) Pixels = $1280 \times 720 = 921,600$. Bits = $921,600 \times 24 = 22,118,400$ bits. Bytes = $22,118,400 / 8 = 2,764,800$ bytes. MiB = $2,764,800 / 1,048,576 = 2.6367...$ MiB, so approximately 2.64 MiB. [2]

3. Sound representation and file size

[4]

- (a) Sample rate is the number of samples taken from the analogue sound wave per second. [1]
(b) Sample resolution / bit depth is the number of bits used to represent the amplitude of each sample. [1]
(c) $22,050 \times 10 \times 16 = 3,528,000$ bits; $3,528,000 / 8 = 441,000$ bytes. [2]

4. Network devices

[5]

No.	Description	Device
1	Forwards packets between different IP networks using destination IP addresses.	Router
2	Connects devices within a LAN and forwards frames to the correct port using MAC addresses.	Switch
3	Converts/modulates and demodulates signals so a network can communicate over an ISP access line.	Modem
4	Connects networks that use different protocols and can perform protocol translation.	Gateway
5	Sends incoming data out to all other ports rather than selecting one destination port.	Hub

Award 1 mark for each correct device.

5. Classful and classless IPv4 addressing / subnetting

[5]

- (a) Class B. **[1]**
- (b) 255.255.0.0. **[1]**
- (c) /26 subnet mask = 255.255.255.192. **[1]**
- (d) Network address = 192.168.10.64. **[1]**
- (e) Broadcast address = 192.168.10.127. **[1]**

Working note: /26 gives blocks of 64 in the final octet: 0-63, 64-127, 128-191, 192-255. Host .70 is in the 64-127 subnet.

6. Cloud computing

[3]

- (a) Accessing files, software, storage or computing services on remote servers over a network / the Internet. **[1]**
- (b) Award 1 mark for one valid benefit, e.g. accessible from different locations with Internet access; reduced need for local storage/software; easier sharing/collaboration; provider may handle backups. **[1]**
- (c) Award 1 mark for one valid drawback, e.g. unavailable without Internet; dependent on a third-party provider/server; possible security/privacy concerns; ongoing service costs. **[1]**

7. Storage hierarchy, swap space and page faults

[3]

- (a) Registers -> cache -> RAM -> secondary storage (SSD/HDD). **[1]**
- (b) Cost per bit generally decreases as storage is farther from the CPU. Access becomes slower / latency increases. **[1]**
- (c) A page fault occurs when the page needed by a process is not currently in RAM. The operating system must fetch the page from secondary storage, commonly the swap/page-file area, and load it into RAM. **[1]**

8. Types of RAM and ROM

[3]

- (a) Award 1 mark for one valid difference: SRAM does not need refreshing and is generally faster/more expensive; DRAM needs periodic refreshing and is generally slower/cheaper. **[1]**
- (b) EPROM (Erasable Programmable Read-Only Memory). **[1]**
- (c) EEPROM (Electrically Erasable Programmable Read-Only Memory). **[1]**